

Wyatt Brown

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Data Science undergraduate (B.S., December 2026) with hands-on technical support and systems work during the degree. Comfortable carrying a project from requirements through handoff: working out what a non-technical stakeholder needs, building and configuring the system, then training the people who run it. Depth in machine learning and probabilistic AI alongside practical work with hardware and small-business networks.

EDUCATION

University of Minnesota — B.S. Data Science, 3.55 GPA

Minneapolis, MN | Expected December 2026

- **Systems & Software:** Practice of Database Systems, Intro to Computer Systems (C, memory hierarchy, processes, IPC), Software Development Process, Intro to Software Development, Discrete Structures & Algorithms
- **AI & Statistics:** Artificial Intelligence II (Bayesian networks, reinforcement learning), Machine Learning Fundamentals, Applied Machine Learning, Operations Research for Data Science, Optimization for Machine Learning (Python, Julia/JuMP), Regression & Statistical Computing, Theory of Statistics I, Applied Multivariate Methods
- **Fall 2026:** Statistical Machine Learning I, Theory of Statistics II, GenAI for Software Engineering, Systems Engineering I

TECHNICAL SKILLS

- **Languages:** Python, R, Java, Julia, TypeScript/JavaScript, SQL
- **Libraries & Frameworks:** JAX (Flax, Optax), PyTorch, Scikit-learn, Pandas, NumPy, React, Flask
- **Infrastructure & Tools:** Git, Postman, Jupyter, Docker, Cloudflare Workers, REST APIs, Vite, QuickBooks, Excel, Adobe Suite
- **Systems & Support:** Hardware and network troubleshooting, wireless and small-network support, security system installation, data migration, end-user training and documentation

PROFESSIONAL EXPERIENCE

Chagos Landscaping LLC — Technical & Operations Consultant

July 2026 – Present

- Sole technical consultant for a family-owned landscaping business with no online presence: built the company's React and TypeScript website with an English/Spanish localization system, and set up its business email, managed phone line, and Google Business Profile.
- Rebuilt the company logo from scratch after the original design files were lost, and wrote the voice-and-tone standards all customer-facing material is checked against.
- Digitized 168 historical invoices and the client and item records the business had no organized copy of; the resulting datasets are its pricing and invoicing reference.
- Produce client estimates and invoices, working directly with the owner to turn what he describes into finished documents.
- Gated the customer request form behind a written security review because it collects customer contact information, and documented the sequence for bringing it live.

XLR8 Sports Training LLC — Freelance IT Consultant & Business Operations

Menasha, WI | October 2022 – January 2025

- Primary onsite IT support for computers, networks, and printers; helped select a security system, then installed and configured it and trained the owner to operate it.
- Migrated 5+ years of business records from paper files to secure digital storage with no loss of billing history.
- Handled scheduling, payment processing (QuickBooks), and onboarding across a client base of 100+, working inside the owner's existing systems and tightening the workflows around them.
- Diagnosed connectivity and hardware failures on site, coordinating with the vendors and staff responsible for each part of the setup.
- Produced graphic design and social media content for the business.

TECHNICAL PROJECTS

Hint Bank — Real-Time Multiplayer Web Game (wrbr0wn.github.io/hintbank)

React, TypeScript, Cloudflare Workers

- Designed a server-authoritative multiplayer architecture (one Durable Object per room) in which the server sends each client only its role-filtered view. Screens that must not see the secret answer never receive it, so an inspected or compromised client has nothing to leak.
- Placed the game rules in a shared protocol module imported by both client and server so the two cannot disagree; clients hold no authoritative state, so reconnection works without any special handling.
- Built an edition-agnostic rules engine with test suites for the client, the engine, and the multiplayer worker in a real workerd runtime.

Pec-King Order — Multi-Agent Reinforcement Learning Under Uncertainty

JAX, Flax, gymnaX

- Built and debugged a 64-agent, 256-tournament simulation comparing tournament-level PPO against MCTS/UCT planning, including rewriting inherited code to reach a working system.
- Implemented Bayesian belief updates by variable elimination over a v-structure Bayes net, letting agents infer hidden opponent strength from observed outcomes alone.
- Swept the planning budget across a 50x range, found no systematic gain, and traced it to a flat reward landscape rather than an implementation fault, shown by uniform UCT visit counts. Confirmed the planner concentrated correctly once exploitable structure existed.

Vending Machine Assortment Optimization

Python (CART), Julia/JuMP

- Wrote the entire modeling and optimization codebase for a predict-then-optimize system covering a 239-machine university fleet: a CART classifier feeding an integer linear program that minimizes expected economic regret.
- Engineered a sales-velocity feature to correct temporal bias, and chose an interpretable tree over higher-scoring ensemble models so operators could see why a product was being pulled.
- Reduced expected opportunity cost roughly 51% per machine across four seasonal scenarios, while filling every slot with no wasted placements.

Also: Dockerized particle simulator (Java engine, Python Flask API, two container configurations isolating the API and the full stack); portfolio site at wyattbrown.info.